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Pivot delivery system for implantable medical device

Abstract

Various aspects of the present disclosure are directed toward apparatuses, systems, and methods that include steering an implantable medical device. The apparatuses, systems, and methods may include an actuation line; a pivot coupled to the implantable medical device; and a tether attached at one end to the actuation line and arranged through the pivot.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a national phase application of PCT Application No. PCT/US2018/049057, internationally filed on Aug. 31, 2018, which is herein incorporated by reference in its entirety for all purposes.

FIELD

(2) The present invention relates to medical devices and methods for treating an anatomical space (e.g., vessels) of the body. More specifically, the invention relates to methods, apparatuses, and systems that include an implantable medical device prosthesis that allows for accurate deployment

in the anatomical space.

BACKGROUND

(3) Disease of the vasculature is increasingly common. Treatment of the vasculature may be difficult because of the tortuous nature and complexity of the vasculature. Aortic dissections, for example, commonly begin at or near the aortic valve root and continue to the ascending aorta and the aortic arch, and may also affect the upper part of the descending aorta. Medical devices implanted at a diseased state may be used for treatment of aortic dissections, aneurysms, and other diseases of the vasculature or other luminal systems of the body, such as the biliary tract, gastrointestinal tract, or respiratory system, for example.

(4) It remains desirable to provide medical devices, systems and methods for repairing disease along the aorta and also for repairing disease along branches extending therefrom.

SUMMARY

(5) According to one example (“Example 1”), a system for steering an implantable medical device includes an actuation line; a pivot coupled to the implantable medical device; and a tether attached at one end to the actuation line and arranged through the pivot and configured to orient the implantable medical device in response to tension applied to the actuation line and release from the pivot after the implantable medical device is oriented.

(6) According to another example (“Example 2”), further to the system of Example 1, the pivot includes a loop attached to an exterior surface of the implantable medical device.

(7) According to another example (“Example 3”), further to the system of Example 2, the loop includes a layer of graft material that forms a lumen between the exterior surface of the implantable medical device and the graft material.

(8) According to another example (“Example 4”), further to the system of Example 3, the actuation line includes an eyelet and the tether is attached to the eyelet at the one end and arranged through the pivot and the eyelet.

(9) According to another example (“Example 5”), further to the system of Example 4, the tether is arranged through the eyelet, arranged through the loop, and subsequently attached to the eyelet.

(10) According to another example (“Example 6”), further to the system of any one of Examples 1-5, the pivot and the actuation line are configured to form a pulley to orient the implantable medical device in response to tension applied to the actuation line.

(11) According to another example (“Example 7”), further to the system of any one of Examples 1-6, the system also includes an actuation line lumen and the tether is pulled into the actuation line lumen in response to tension applied to the actuation line.

(12) According to another example (“Example 8”), further to the system of Example 7, the actuation line lumen is attached to the exterior surface of the implantable medical device proximal to the pivot.

(13) According to another example (“Example 9”), further to the system of any one of Examples 7-8, the pivot and the actuation line lumen are configured to form the pulley to orient the implantable medical device in response to tension applied to the actuation line.

(14) According to another example (“Example 10”), further to the system of any one of Examples 1-9, further comprising a removable lock wire configured to maintain coupling of the tether to the implantable medical device.

(15) According to another example (“Example 11”), further to the system of Example 10, the tether includes an eyelet and the removable lock wire is arranged through the eyelet to couple the removable lock wire to the tether.

(16) According to another example (“Example 12”), further to the system of Example 11, the eyelet of the tether is at or near a proximal end of the tether and a distal end of the tether is coupled to the actuation line.

(17) According to another example (“Example 13”), further to the system of Example 12, the removable lock wire is arranged through a flow lumen of the implantable medical device and the

tether is arranged from the flow lumen of the implantable medical device at the proximal end of the tether to an exterior surface of the removable lock wire at the distal end of the tether.

(18) According to another example (“Example 14”), further to the system of Example 13, the tether is configured to release from the actuation line in response to withdraw of the removable lock wire from the eyelet of the tether.

(19) According to another example (“Example 15”), further to the system of Example 10, the system also includes a catheter arranged through a lumen of the implantable medical device, wherein a proximal end of the tether is releasably coupled to the catheter, and a distal end of the tether is coupled to the actuation line.

(20) According to an example (“Example 16”) a delivery system includes a catheter; an implantable medical device arranged near a leading end of the catheter and including a proximal end, a distal end, and a flow lumen extending therebetween; a loop coupled to an exterior surface of the implantable medical device; a lumen coupled to the exterior surface of the implantable medical device proximal to the loop; an actuation line arranged through the lumen; and a tether coupled to the actuation line and arranged through the loop and configured to steer the implantable medical device in response to tension applied to the actuation line.

(21) According to another example (“Example 17”), further to the system of Example 16, the loop and the lumen are pivot points and are configured to form a pulley between the tether and the actuation line to steer the implantable medical device in response to tension applied to the actuation line.

(22) According to another example (“Example 18”), further to the system of Example 16, the system also includes a removable lock wire configured to maintain coupling of the tether to the implantable medical device, and wherein the tether is configured to release from the actuation line in response to withdraw of the removable lock wire.

(23) According to another example (“Example 19”), a method of steering an implantable medical device includes delivering the implantable medical device to a target location within a patient's vasculature; and manipulating an actuation line, coupled to the implantable medical device by a tether arranged through a loop coupled to an exterior surface of the implantable medical device, to steer the implantable medical device.

(24) According to another example (“Example 20”), further to the method of Example 19, the method also includes maintaining coupling of the tether to the implantable medical device by a removable lock wire, and wherein releasing the tether from the loop in response to withdraw of the removable lock wire.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate embodiments, and together with the description serve to explain the principles of the disclosure.

(2) FIG. 1 shows an implantable medical device and an actuation line in accordance with an embodiment.

(3) FIG. 2 shows another implantable medical device and an actuation line in accordance with an embodiment.

(4) FIG. 3A shows an illustration of an exterior view of an example delivery system in an unsteered configuration in accordance with an embodiment.

(5) FIG. 3B shows an illustration of an exterior view of the delivery system, shown in FIG. 3A, in a steered configuration in accordance with an embodiment.

(6) FIG. 4A shows an illustration of an exterior view of another example delivery system in an

unsteered configuration in accordance with an embodiment.

(7) FIG. 4B shows an illustration of an interior view of the delivery system, shown in FIG. 4A, in accordance with an embodiment.

(8) FIG. 4C shows an illustration of an exterior view of the delivery system, shown in FIGS. 4A-4B, in a steered configuration in accordance with an embodiment.

(9) FIG. 4D shows an illustration of an interior view of the delivery system, shown in FIGS. 4A-C, in a configuration for removal of the tether in accordance with an embodiment.

(10) FIG. 4E shows an illustration of an exterior view of the delivery system, shown in FIGS. 4A-4D, in a configuration for removal of the tether in accordance with an embodiment.

(11) FIG. 5 shows an illustration of an interior view of an implantable device with an example lock wire arrangement in accordance with an embodiment.

(12) FIG. 6A shows an illustration of an exterior view of another example delivery system in accordance with an embodiment.

(13) FIG. 6B shows an illustration of an interior view of the delivery system, shown in FIG. 6A, in accordance with an embodiment.

(14) FIGS. 7A-7E show side view illustrations of expandable device angulation relative to a target location in accordance with various aspects of the present disclosure.

DETAILED DESCRIPTION

(15) Persons skilled in the art will readily appreciate that various aspects of the present disclosure can be realized by any number of methods and apparatus configured to perform the intended functions. It should also be noted that the accompanying drawing figures referred to herein are not necessarily drawn to scale, but may be exaggerated to illustrate various aspects of the present disclosure, and in that regard, the drawing figures should not be construed as limiting.

(16) Various aspects of the present disclosure are directed toward apparatuses, systems, and methods that include an implantable medical device that may be used in treatment of the vasculature. The implantable medical device is delivered to the vasculature using a delivery system. In addition, the implantable medical devices described herein may be substantially cylindrical, include a bifurcation, or any combination of fenestrations. Further, the implantable medical devices may be configured to conform to the vasculature into which the implantable medical device is implanted, low-profile in order to enable delivery thereof using a minimally invasive procedure (e.g., transcatheter), and withstand forces and other stresses that occur once implanted in the vasculature.

(17) The delivery system may be configured to position and/or steer the implantable medical device for accurate placement in the vasculature. To position and or steer the implantable medical device, the delivery system may include an actuation line (e.g., a wire, tether, or other member) that changes the position of the implantable medical device in response to a user applying force to the actuation line. The actuation line may be releasably coupled to the implantable medical device to avoid trauma to the vasculature after the implantable medical device is delivered and positioned. As discussed in further detail below, a tether, coupled to the actuation line, may be used in combination with a pivot coupled to the implantable medical device to orient the implantable medical device in response to tension applied to the actuation line. In certain instances, the actuation line, pivot, and tether may form a pulley arrangement that facilitates orienting of the implantable medical device with a patient's tortuous anatomy.

(18) FIG. 1 shows an implantable medical device **100** and an actuation line **102** in accordance with various aspects of the present disclosure. The implantable medical device **100** is releasably coupled to a delivery system for delivery of the implantable medical device **100** to a target location within a patient's vasculature. The delivery system may include a catheter **104** that includes a leading end **106** and a trailing end (not shown in FIG. 1). The implantable medical device **100** may be arranged near the leading end **106** of the catheter **104**. The catheter **104** may extend through a lumen of the implantable medical device **100** toward and past a proximal end **108** of the implantable medical

device **100**. The catheter **104** may also include a tip (not shown) at the leading end **106**. As shown in FIG. 1, the implantable medical device **100** includes stent and graft components.

(19) The implantable medical device **100** includes a proximal end **108**, a distal end **110**, and a flow lumen extending therebetween. The proximal end **108** of the implantable medical device **100** may be considered the end of the implantable medical device **100** that is closest to the target location within the patient's vasculature. The actuation line **102** may be coupled to the implantable medical device **100** at one or more locations. As shown in FIG. 1, the actuation line **102** is attached adjacent to or near the proximal end **108** of the implantable medical device **100** and accessible to a user of the delivery system. The actuation line **102** may be attached to other portions of the implantable medical device **100**.

(20) As shown, the actuation line **102** is coupled to the implantable medical device **100** via at least one tether **112**. The tether **112** may be arranged through a portion of the implantable medical device **100** and through the actuation line **102** to couple the line **102** to the implantable medical device **100**. In certain instances, and as shown in FIG. 1, the at least one tether **112** is arranged through the implantable medical device **100** near or adjacent to the proximal end **108** of the implantable medical device **100**. The at least one tether **112** may be a single tether, as shown in FIG. 1. In other instances, the implantable medical device **100** may include a loop **114** attached or coupled to an exterior surface of the implantable medical device **100**. The loop **114** may include a layer of graft material that forms a lumen between the exterior surface of the implantable medical device **100** and the graft material. The tether **112** may be arranged through the loop **114** to couple the line **102** to the implantable medical device **100**. In other instances, the loop **114** or pivot may be a hole or holes through graft material of the implantable medical device **100**. In these instances, the tether **112** is arranged through the hole or holes of implantable medical device **100**.

(21) In certain instances, the line **102** is an actuation line **102** configured to steer/orient the implantable medical device **100** during delivery thereof. The actuation line **102** may include a stiffness such that a user operating the delivery system may apply force to the actuation line **102** and bidirectionally steer (e.g., proximally and distally relative to the target location within the patient's vasculature) the implantable medical device **100**. For example, the actuation line **102** may have a stiffness that is greater than a stiffness of the tether **112**. The stiffness of the actuation line **102** and/or the location to which the actuation line **102** is coupled to the implantable medical device **100** may facilitate deploying and arranging the implantable medical device **100** relative to the target location within the patient's vasculature. For example, the implantable medical device **100** may be configured to deploy at a tortuous vessel having a curvature with at least one inflection point. In certain instances, the actuation line **102** is configured to maintain the proximal end **108** of the implantable medical device **100** approximately perpendicular to the inflection point in the curvature of the tortuous vessel during delivery of the implantable medical device **100**.

(22) The actuation line **102** may be uncoupled or released from the implantable medical device **100** subsequent to the implantable medical device **100** being positioned and deployed at the target location within the patient's vasculature and removed from the patient. The actuation line **102** can include an eyelet or opening in a leading end through which the tether **112** is arranged. In addition, the tether **112** may be attached to the actuation line **102** at the eyelet or at another location as explained in further detail below. In other instances, the tether **112** is configured to be removed or unthreaded to uncouple the actuation line **102** from the implantable medical device **100**.

(23) FIG. 2 shows another implantable medical device **100** and an actuation line **102** in accordance with various aspects of the present disclosure. The implantable medical device **100** may be releasably coupled to a delivery system. The delivery system may include a catheter **104** that includes a leading end **106** and a trailing end (not shown in FIG. 2). The implantable medical device **100** may be arranged near the leading end **106** of the catheter **104**. The delivery system may be configured to deliver the implantable medical device **100** to a target location within a patient's vasculature. In certain instances, the implantable medical device **100** may be configured to deploy

at a tortuous vessel having a curvature with at least one inflection point. To facilitate deploying of the implantable medical device **100**, the delivery system may include the actuation line **102** configured to maintain a proximal end **108** (or distal end **110**) of the implantable medical device **100** approximately perpendicular to the inflection point in the curvature of the tortuous vessel during delivery of the implantable medical device **100**.

(24) The actuation line **102** (accessible to a user of the delivery system), for example, is configured to steer/orient the implantable medical device **100** during delivery thereof, and is releasably coupled to the implantable medical device **100** via at least one tether **112**. The tether **112** may be arranged through a portion of the implantable medical device **100** and through the actuation line **102** to couple the actuation line **102** to the implantable medical device **100**. In certain instances, and as shown in FIG. 2, the at least one tether **112** is arranged through the implantable medical device **100** near or adjacent to the proximal end **108** of the implantable medical device **100**. In other instances, the implantable medical device **100** may include a loop **114** attached or coupled to an exterior surface of the implantable medical device **100**. The loop **114** may include a layer of graft material that forms a lumen between the exterior surface of the implantable medical device **100** and the graft material. The tether **112** may be arranged through the loop **114** to couple the line **102** to the implantable medical device **100**. In other instances, the loop **114** or pivot may be a hole or holes through graft material of the implantable medical device **100**. In these instances, the tether **112** is arranged through the hole or holes of implantable medical device **100**.

(25) In addition, the actuation line **102** may be arranged through a sleeve **214** that is attached to an exterior portion of the implantable medical device **100**. The implantable medical device **100** may include a graft component and one or more stent components. The sleeve **214** may be formed of a similar material or the same material as the graft component of the implantable medical device **100**. The sleeve **214** may include a lumen through which the actuation line **102** is arranged. In certain instances, the sleeve **214** is an enclosed structure which forms the lumen, or the sleeve **214** is a layer of graft material that forms a lumen between the sleeve **214** and the implantable medical device **100**. The sleeve **214** may facilitate the actuation line **102** steering the implantable medical device **100**. The sleeve **214** may prevent traumatic interaction between the actuation line **102** and a vessel wall. In addition, the sleeve **214** may enhance the connection between the actuation line **102** and the implantable medical device **100** when a user applies force or tension to the actuation line **102**. As shown, the sleeve **214** has a length similar to the length of the implantable medical device **100**. In other instances, the sleeve **214** may have a shorter length than the implantable medical device **100** or a longer length than the implantable medical device.

(26) The actuation line **102** may include a stiffness such that a user operating the delivery system may apply force to the actuation line **102** and bidirectionally steer (e.g., proximally and distally relative to the target location within the patient's vasculature) the implantable medical device **100**. For example, the actuation line **102** may have a stiffness that is greater than a stiffness of the tether **112**. The stiffness of the actuation line **102** and/or the location to which the actuation line **102** is coupled to the implantable medical device **100** may facilitate deploying and arranging the implantable medical device **100** relative to the target location within the patient's vasculature.

(27) FIG. 3A shows an illustration of an exterior view of an example delivery system **300** in an unsteered configuration in accordance with an embodiment. The delivery system **300** may be used for steering an implantable medical device **100** (e.g., as shown in FIGS. 1-2). The delivery system **300** may include an actuation line **102**. As shown in FIGS. 3A-B, the actuation line **102** includes an eyelet **306**. In addition, the delivery system **300** also includes a pivot **114** and a tether **112**. For ease of illustration, element **308** is illustrative of a hole in the implantable medical device **100**. As FIGS. 3A-B are an illustration from an exterior view of the implantable medical device **100**, the element **308** indicates a pathway from an exterior surface of the implantable medical device **100** to an interior (flow lumen) of the implantable medical device **100**.

(28) Consistent with the loop shown in FIGS. 1-2, the pivot **114** may be coupled to the implantable

medical device **100**. In addition, and as shown in FIGS. 1-2, the pivot **114** may also be attached to an exterior surface of the implantable medical device **100**. In certain instances, the pivot **114** may be a hole or holes in the implantable medical device **100**. In other instances, the pivot **114** is a loop that includes a layer of graft material that forms a lumen between the exterior surface of the implantable medical device **100** and the graft material.

(29) As shown in FIGS. 3A-B, the tether **112** may be arranged through the loop **114** to couple the actuation line **102** to the implantable medical device **100**. In addition, one end of the tether **112** may be attached to the actuation line **102**. In certain instances, the tether **112** may be attached to the actuation line **102** at the eyelet **306**. From the attachment at the eyelet **306** (or another distal end portion of the actuation line **102**), the tether **112** is arranged through the pivot **114**. In addition, after being arranged through the pivot **114** (or loop), the tether **112** travels proximally (e.g., toward a user along an exterior surface of the implantable medical device **100**).

(30) In certain instances, each of the tether **112** and the actuation line **102** may transition between the exterior surface of the implantable medical device **100** to an interior (flow lumen) of the implantable medical device **100** at element **308**. At this point, the tether **112** and the actuation line **102** may enter a catheter (e.g., as shown in FIGS. 1-2) at which the implantable medical device **100** may be arranged for delivery. The tether **112** and/or the actuation line **102** can travel proximally toward a user or a handle portion of the delivery system **300**. In this manner, a user may apply tension to the tether **112** and/or the actuation line **102**. FIG. 3B shows an example illustration of the delivery system **300** in a steering configuration (e.g., when a user has applied tension to the actuation line **102**. As shown in FIG. 3B, the actuation line **102** has been withdrawn proximally relative to the position in the unsteered configuration shown in FIG. 3A.

(31) The tether **112**, by way of attachment or coupling to the actuation line **102**, is pulled or drawn proximally along with the actuation line **102** when tension is applied to the actuation line **102**. As a result of the tether **112** being arranged through the pivot **114**, the tether **112** will curve, orient, or actuate the implantable medical device **100**. The pivot **114** being coupled or attached to the implantable medical device **100** and the tether **112** being arranged through the pivot **114** pulls, curves, orients, or forces a configuration change to an end of the implantable medical device **100** to which the pivot **114** is coupled. As shown in FIG. 7A, the actuation line **102**, tether **112**, and pivot **114** may, in combination, angle or curve a leading end of the device in tortuous anatomy to orient the implantable medical device **100**.

(32) In certain instances, the pivot **114** and the actuation line **102** are configured to form a pulley to orient the implantable medical device **100** in response to tension applied to the actuation line **102**. The tether **112** may loop around the pivot **114** to form a pulley when the actuation line **102** is tensioned or pulled proximally.

(33) FIG. 4A shows an illustration of an exterior view of another example delivery system **400** in an unsteered configuration in accordance with an embodiment. The delivery system **400** may be used for steering an implantable medical device **100** (e.g., as shown in FIGS. 1-2). The delivery **400** may include an actuation line **102** and an actuation line sleeve **214**. The actuation line sleeve **214** may be attached or coupled to an exterior surface of the implantable medical device **100**. In addition, the actuation line sleeve **214** may facilitate the actuation line **102** steering the implantable medical device **100**. The actuation line sleeve **214** may prevent traumatic interaction between the actuation line **102** and a vessel wall (and may form a part of a pulley as discussed in further detail below). In addition, the actuation line sleeve **214** may enhance the connection between the actuation line **102** and the implantable medical device **100** when a user applies force or tension to the actuation line **102**. The delivery system **400** may also include a pivot or loop **114** coupled to the implantable medical device. The loop **114** may include a layer of loop material that forms a lumen between the exterior surface of the implantable medical device **100** defined by a graft material. The tether **112** may be arranged through the loop **114** and external to the actuation line sleeve **214**.

(34) As shown in FIG. 4A, the actuation line **102** includes an eyelet **306**. One end of the tether **112**

may be attached to the actuation line **102**. In certain instances, the tether **112** may be attached to the actuation line **102** at the eyelet **306**. From the attachment at the eyelet **306** (or another distal end portion of the actuation line **102**), the tether **112** is arranged through the loop **114**. In addition, after being arranged through the loop **114** (or pivot), the tether **112** travels proximally (e.g., toward a user along an exterior surface of the implantable medical device **100**). As shown in FIG. **4A**, the tether **112** is arranged through the eyelet **306**, arranged through the loop **114**, and subsequently attached to the eyelet **306** (e.g., as the tether **112** approaches a leading or distal end of the implantable medical device **100**).

(35) In certain instances, each of the tether **112** and the actuation line **102** may transition between the exterior surface of the implantable medical device **100** to an interior (flow lumen) of the implantable medical device **100** at element **308**. For ease of illustration, element **308** is illustrative of a hole in the implantable medical device **100**. FIG. **4A** is an illustration from an exterior view of the implantable medical device **100** whereas FIG. **4B** is an illustration from an interior view of the implantable medical device **100** with the element **308** indicates a pathway from an exterior surface of the implantable medical device **100** to an interior (flow lumen) of the implantable medical device **100**.

(36) As shown in FIG. **4B**, the delivery system **400** may include a removable lock wire **420**. The removable lock wire **420**, in certain instances, is configured to maintain coupling of the tether **112** to the implantable medical device **100**. In addition, the tether **112** is configured to release from the actuation line **102** in response to withdrawal of the removable lock wire **420** as shown in FIGS. **4D-E** and explained in further detail below. In certain instances, the removable lock wire **420** may be attached to the tether **112**. In other instances, and as shown in FIG. **4B**, the removable lock wire **420** may be arranged through a portion of the tether **112**. The tether **112**, for example, may include an eyelet **422** arranged at an end of the tether **112**. In instances where the tether **112** includes an eyelet **422**, the removable lock wire **420** is arranged through the eyelet **422** of the tether **112** to couple the removable lock wire **420** to the tether **112**.

(37) The eyelet **422** of the tether **112** may be arranged at one end of the tether **112** with the other end of the tether **112** being attached to the eyelet **306** of the actuation line **102**. In certain instances, the eyelet **422** of the tether **112** is at or near a proximal end of the tether **112** and a distal end of the tether **112** is coupled to the actuation line **102**.

(38) As shown in FIG. **4B**, the removable lock wire **420** is arranged through a flow lumen of the implantable medical device **100** and the tether **112** is arranged from the flow lumen (e.g., interior portion) of the implantable medical device **100** at the proximal end of the tether **112** to an exterior surface of the removable lock wire **420** at the distal end of the tether.

(39) As also shown in FIG. **4B**, the delivery system **400** may include a catheter **104** with the implantable medical device **100** arranged near a leading end of the catheter **104** (as shown in further detail in FIGS. **1-2**). The catheter **104** may extend through a lumen of the implantable medical device **100** toward and past a leading end of the implantable medical device **100**. The catheter **104** may also include a tip (not shown) at the leading end. The removable lock wire **420** may be arranged through a lumen of the catheter **104**, and may be withdrawn into the lumen of the catheter **104**, as shown in FIG. **4D**, to release the tether **112**.

(40) FIG. **4C** shows an illustration of an exterior view the delivery system **400**, shown in FIGS. **4A-B**, in a steered configuration in accordance with an embodiment. The tether **112**, by way of attachment or coupling to the actuation line **102**, is pulled or drawn proximally along with the actuation line **102** when tension is applied to the actuation line **102**. As a result of the tether **112** being arranged through the loop **114**, the tether **112** will curve, orient, or actuate the implantable medical device **100**. The loop **114** being coupled or attached to the implantable medical device **100** and the tether **112** being arranged through the loop **114** pulls, curves, orients, or forces a configuration change to an end of the implantable medical device **100** to which the loop **114** is coupled. As shown in FIG. **7A**, the actuation line **102**, tether **112**, and loop **114** may, in

combination, angle or curve a leading end of the device in tortuous anatomy to orient the implantable medical device **100**.

(41) In addition, the tether **112** is pulled into a lumen defined by the actuation line sleeve **214** in response to tension applied to the actuation line **102**. In certain instances, the tether **112** may follow a pathway that begins from the tether **112** attachment at the eyelet **306** of the actuation line **102**, the tether **112** loops through the loop **114** back into the lumen defined by the actuation line sleeve **214**, through the eyelet **306** of the actuation line **102**, and back out of the lumen defined by the actuation line sleeve **214** as shown in FIG. 4C. The tether **112** (and actuation line **102**) may then enter the implantable medical device **100** at element **308**. At this point, the actuation line **102** may enter the catheter **104** and travel proximally toward a user or a handle portion of the delivery system **400**. In this manner, a user may apply tension to the actuation line **102**.

(42) As shown in FIG. 4C, the actuation line sleeve **214** is attached to an exterior surface of the implantable medical device **100** proximal to the loop **114** (or pivot). In certain instances, the pathway shown in FIG. 4C forms a pulley arrangement for orienting the implantable medical device **100**. For example, the loop **114** and the actuation line sleeve **214** are configured to form a pulley to orient the implantable medical device **100** in response to tension applied to the actuation line **102**. The loop **114** and the actuation line sleeve **214** are pivot points in the pulley and are configured to form a pulley between the tether **112** and the actuation line **102** to steer the implantable medical device **100** in response to tension applied to the actuation line **102**.

(43) FIG. 4D shows an illustration of an interior view of the delivery system **400** in a configuration for removal of the tether **112**. The tether **112** is configured to release from the actuation line **102** in response to withdrawal of the removable lock wire **420** from the eyelet **422** of the tether **112**. As shown in FIG. 4D, the removable lock wire **420** may be withdrawn into the catheter **104**. The removable lock wire **420** may be removed after the implantable medical device **100** is oriented in a desired configuration at a target location within a patient.

(44) FIG. 4E shows an illustration of an exterior view of the delivery system **400** in a configuration for removal of the tether in accordance with an embodiment. The tether **112** is configured to release from the actuation line **102** in response to withdraw of the removable lock wire **420**. Continued tension applied to the actuation line **102** after release or removal of the removable lock wire **420** pulls the tether **112** through element **308** (e.g., a hole in the implantable medical device **100**). At this point, the tether **112** and the actuation line **102** are uncoupled from the medical device **100**. In addition, a user may continue to apply tension or withdraw the actuation line **102** to remove the actuation line **102** and the tether **112**, attached to the actuation line **102**, from the patient.

(45) FIG. 5 shows an illustration of an interior view of an implantable device **100** with an example lock wire **420** arrangement in accordance with an embodiment. A catheter **104** is also shown in FIG. 5, the catheter **104** is arranged through a flow lumen of the implantable medical device **100**. At a distal end of the catheter **104** is an olive **524**. The olive **524** may be an atraumatic tip of the catheter **104** and delivery system.

(46) In certain instances, a lock wire **420**, as discussed in further detail above with reference to FIGS. 4A-E, may be embedded in a portion of the olive **524**. The olive **524** may include a cut-away section **526** as shown in FIG. 5. In certain instances, the lock wire **420** is embedded in the cut-away section **526** of the olive **524** during delivering and steering/orienting of the implantable medical device **100**. The lock wire **420** may be arranged through an eyelet **422** of a tether **112** for steering/orienting of the implantable medical device **100** as discussed in detail above. An end of the tether **112** that includes the eyelet **422** is coupled to the catheter **104** by way of the lock wire **420** with another end of the tether **112** being external to the implantable medical device **100** for attachment or coupling to an actuation line **102** (not shown). As shown in FIG. 5, the tether **112** transitions from an interior of the implantable medical device **100** to an exterior by being arranged through element **308** (e.g., a hole in the implantable medical device **100**).

(47) The catheter **104** shown in FIG. 5 is provided as an example of the various features of the

catheter **104** and, although the combination of those illustrated features is clearly within the scope of invention, that example and its illustration is not meant to suggest the inventive concepts provided herein are limited from fewer features, additional features, or alternative features to one or more of those features shown in FIG. 5. For example, in various embodiments, the catheter **104** and/or lock wire **420** arrangement shown in FIG. 5 may be included in the delivery systems described with reference to FIGS. 1-4. The delivery systems, for example, may include an olive **524** or a lock wire **420** may be pinned to the olive **524**. It should also be understood that the reverse is true as well. For example, the tether **112** and lock wire **420** shown in FIG. 5 may be employed in connection with the actuation line **102** shown in FIGS. 4A-E.

(48) FIG. 6A shows an illustration of an exterior view of another example delivery system **600** in accordance with an embodiment. The delivery system **600** may be used for steering an implantable medical device (e.g., as shown in FIGS. 1-2). The delivery system **600** may include an actuation line **102** and a sleeve **214**. The sleeve **214** may be attached or coupled to an exterior surface of the implantable medical device **100**.

(49) The delivery system **600** may also include a pivot or loop **114** coupled to the implantable medical device **100**. The loop **114** may include a layer of loop material that forms a lumen between the exterior surface of the implantable medical device **100** and defined by a graft material. The tether **112** may be arranged through the loop **114** and external to the sleeve **214**. As shown in FIG. 6A, the actuation line **102** includes an eyelet **306** with the tether **112** being routed through the eyelet **306**. In addition, and as shown in FIG. 6A, the tether **112** includes two portions **626**, **628** as a result of the looping through the eyelet **306** of the actuation line **102**. The portions **626**, **628** of the tether **112** are arranged through the lumen defined by the sleeve **214** and routed to an interior of the implantable medical device **100** through a hole in the implantable medical device **100** illustrated by element **308**.

(50) FIG. 6B shows an illustration of an interior view of the delivery system **600**, shown in FIG. 6A, in accordance with an embodiment. One of the portions **626** of the tether **112** may be attached to a catheter as illustrated by element **630**. In addition, the other of the portions **628** may be coupled to a portion of a delivery handle **632**, accessible to a user. The user may apply tension to remove the portion of the delivery handle **632** to apply tension to the tether **112**. The tether **112** may then be released from the attachment point **630** on the catheter and thereby decouple from the eyelet **306** of the actuation line **102**. This decouples the actuation line **102** from the implantable medical device **100** after orientation is accomplished.

(51) FIGS. 7A-E show side view illustrations of expandable device angulation relative to a target location **700a-e** in accordance with various aspects of the present disclosure. Each of FIGS. 7A-E show a side profile of a leading (or proximal) end **700a-e** of an expandable device, consistent with various aspects of the present disclosure. In certain instances, the target location **700a-e** may be at a tortuous vessel of a patient. The target location **700a-e** into which the expandable device is implanted may have angulation (e.g., a curvature with at least one inflection point **704a-e**). The target location **700a-e** may be an angulated abdominal aortic aneurism (AAA).

(52) In certain instances, one of the ends **702a-e** of the expandable device may be deployed perpendicular to the inflection point in the curvature of the tortuous vessel during delivery of the expandable device. Non-perpendicularity may negatively affect the ability of the expandable device to seal against the target location **700a-e**. FIG. 7A shows the leading (or proximal) end **702a** deployed perpendicular to the inflection point **704a**. In certain instances, perpendicularity of the expandable device may be a function of device flatness, angulation, and rotational alignment. FIG. 7B shows the leading (or proximal) end **702b** of an expandable device angled relative to the inflection point **704b** of the target location **700b**. FIG. 7C shows the leading (or proximal) end **702c** of an expandable device rotated relative to the inflection point **704c** of the target location **700c**. FIG. 7D shows the leading (or proximal) end **702d** of an expandable device deformed relative to the inflection point **704b** of the target location **700d**. FIG. 7E shows the leading (or proximal) end

702e of an expandable device deformed or flat, rotated, and angled relative to the inflection point **704e** of the target location **700e**.

(53) Device deployment and performance can be enhanced by steering the device to an appropriate location while maintaining one of the ends of the expandable device perpendicular to the target location **700a-e** (e.g., curvature of a vessel with at least one inflection point **704a-e**) during and after deployment. The actuation lines and arrangements thereof discussed herein facilitate maintaining the expandable device perpendicular during and after deployment (as shown in FIG. 7A) and mitigate against non-perpendicular, angled, or flat deployment (as shown in FIGS. 7B-E).

(54) The lines discussed herein may be formed from metallic, polymeric or natural materials such as stainless steels, cobalt-chromium alloys and nitinol. Further, actuation lines can also be formed from high strength polymer fibers such as ultra high molecular weight polyethylene fibers (e.g., Spectra™, Dyneema Purity™, etc.) or aramid fibers (e.g., Technora™, etc.). In certain instances, the actuation line may have a great column strength than the tethers.

(55) The graft components may be made up of any material which is suitable for use as a graft in the chosen body lumen and being resistant to expansion as discussed herein. The graft components may be composed of the same or different materials. Furthermore, the graft components may include multiple layers of material that can be the same material or different material. In one embodiment, said materials can be used in combination and assembled together to comprise a graft. The graft materials used in a stent graft can be extruded, coated or formed from wrapped films, or a combination thereof. Polymers, biodegradable and natural materials can be used for specific applications.

(56) Examples of synthetic polymers include, but are not limited to, nylon, polyacrylamide, polycarbonate, polyformaldehyde, polymethylmethacrylate, polytetrafluoroethylene, polytrifluorochlorethylene, polyvinylchloride, polyurethane, elastomeric organosilicon polymers, polyethylene, polypropylene, polyurethane, polyglycolic acid, polyesters, polyamides, their mixtures, blends and copolymers are suitable as a graft material. In one embodiment, said graft is made from a class of polyesters such as polyethylene terephthalate including DACRON® and MYLAR® and polyaramids such as KEVLAR®, polyfluorocarbons such as polytetrafluoroethylene (PTFE) with and without copolymerized hexafluoropropylene (TEFLON® or GORE-TEX®), and porous or nonporous polyurethanes. In another embodiment, said graft comprises expanded fluorocarbon polymers (especially PTFE) materials. Included in the class of preferred fluoropolymers are polytetrafluoroethylene (PTFE), fluorinated ethylene propylene (FEP), copolymers of tetrafluoroethylene (TFE) and perfluoro(propyl vinyl ether) (PFA), homopolymers of polychlorotrifluoroethylene (PCTFE), and its copolymers with TFE, ethylene-chlorotrifluoroethylene (ECTFE), copolymers of ethylene-tetrafluoroethylene (ETFE), polyvinylidene fluoride (PVDF), and polyvinylfluoride (PVF). Especially preferred, because of its widespread use in vascular prostheses, is ePTFE. In another embodiment, said graft comprises a combination of said materials listed above. In another embodiment, said graft is substantially impermeable to bodily fluids. Said substantially impermeable graft can be made from materials that are substantially impermeable to bodily fluids or can be constructed from permeable materials treated or manufactured to be substantially impermeable to bodily fluids (e.g. by layering different types of materials described above or known in the art). In another embodiment, said outermost tube comprises ePTFE. In another embodiment, said innermost tube comprises ePTFE. In another embodiment, said innermost and outermost tube comprises ePTFE film that has been wrapped into a tube. In another embodiment, said secondary stent is covered with any of the material disclosed herein or known in the art. In another embodiment, the secondary stent covering comprises ePTFE.

(57) Additional examples of graft materials include, but are not limited to, vinylidene fluoride/hexafluoropropylene hexafluoropropylene (HFP), tetrafluoroethylene (TFE), vinylidene fluoride, 1-hydropentafluoropropylene, perfluoro(methyl vinyl ether), chlorotrifluoroethylene (CTFE), pentafluoropropene, trifluoroethylene, hexafluoroacetone,

hexafluoroisobutylene, fluorinated poly(ethylene-co-propylene (FPEP), poly(hexafluoropropene) (PHFP), poly(chlorotrifluoroethylene) (PCTFE), poly(vinylidene fluoride (PVDF), poly(vinylidene fluoride-co-tetrafluoroethylene) (PVDF-TFE), poly(vinylidene fluoride-co-hexafluoropropene) (PVDF-HFP), poly(tetrafluoroethylene-co-hexafluoropropene) (PTFE-HFP), poly(tetrafluoroethylene-co-vinyl alcohol) (PTFE-VAL), poly(tetrafluoroethylene-co-vinyl acetate) (PTFE-VAC), poly(tetrafluoroethylene-co-propene) (PTFEP) poly(hexafluoropropene-co-vinyl alcohol) (PHFP-VAL), poly(ethylene-co-tetrafluoroethylene) (PETFE), poly(ethylene-co-hexafluoropropene) (PEHFP), poly(vinylidene fluoride-co-chlorotrifluoroethylene) (PVDF-CTFE), and combinations thereof, and additional polymers and copolymers described in U.S. Publication 2004/0063805, incorporated by reference herein in its entirety for all purposes. Additional polyfluorocopolymers include tetrafluoroethylene (TFE)/perfluoroalkylvinylether (PAVE). PAVE can be perfluoromethylvinylether (PMVE), perfluoroethylvinylether (PEVE), or perfluoropropylvinylether (PPVE), as described in U.S. Publication 2006/0198866 and U.S. Pat. No. 7,049,380, both of which are incorporated by reference herein for all purposes in their entireties. Other polymers and copolymers include, polylactide, polycaprolacton-glycolide, polyorthoesters, polyanhydrides; poly-aminoacids; polysaccharides; polyphosphazenes; poly(ether-ester) copolymers, e.g., PEO-PLLA, or blends thereof, polydimethyl-siloxane; poly(ethylene-vingylacetate); acrylate based polymers or copolymers, e.g., poly(hydroxyethyl methylmethacrylate, polyvinyl pyrrolidinone; fluorinated polymers such as polytetrafluoroethylene; cellulose esters and any polymer and copolymers described in U.S. Publication 2004/0063805, incorporated by reference herein in its entity.

(58) The graft components, as discussed herein, may be attached to the self-expanding stent elements by using a coupling member that is generally a flat ribbon or tape having at least one generally flat surface. In certain instances, the tape member is made from expanded PTFE (ePTFE) coated with an adhesive. The adhesive may be a thermoplastic adhesive. In certain instances, the thermoplastic adhesive may be fluorinated ethylene propylene (FEP). More specifically, an FEP-coated side of the ePTFE may face toward and contacts an exterior surface of the self-expanding stent and graft component, thus attaching the self-expanding stent to the graft component.

(59) The stent component(s) discussed herein can be fabricated from a variety of biocompatible materials. These materials may include 316L stainless steel, cobalt-chromium-nickel-molybdenum-iron alloy ("cobalt-chromium"), other cobalt alloys such as L605, tantalum, Nitinol, or other biocompatible metals. In certain instances, as discussed in detail above, the stent (and graft) may be self-expanding. In other instances, the prosthesis may be balloon expandable.

(60) The stent component(s) discussed herein may be constructed from a reasonably high strength material, i.e., one which is resistant to plastic deformation when stressed. In one embodiment, the stent component(s) comprise a wire which is helically wound around a mandrel having pins arranged thereon so that the helical turns and undulations can be formed simultaneously. Other constructions may also be used. In certain instances, the stent component(s) are made from a super-elastic alloy. There are a variety of disclosures in which super-elastic alloys such as nitinol are used in stents. See for example, U.S. Pat. No. 4,503,569, to Dotter; U.S. Pat. No. 4,512,338, to Balko et al.; U.S. Pat. No. 4,990,155, to Wilkoff; U.S. Pat. No. 5,037,427, to Harada, et al.; U.S. Pat. No. 5,147,370, to MacNamara et al.; U.S. Pat. No. 5,211,658, to Clouse; and U.S. Pat. No. 5,221,261, to Termin et al.

(61) A variety of materials variously metallic, super elastic alloys, such as Nitinol, are suitable for use in the stent component(s). Primary requirements of the materials are that they be suitably springy even when fashioned into very thin sheets or small diameter wires. Various stainless steels which have been physically, chemically, and otherwise treated to produce high springiness are suitable as are other metal alloys such as cobalt chrome alloys (e.g., ELGILOY®), platinum/tungsten alloys, and especially the nickel-titanium alloys generically known as "nitinol".

(62) The invention of this application has been described above both generically and with regard to

specific embodiments. It will be apparent to those skilled in the art that various modifications and variations can be made in the embodiments without departing from the scope of the disclosure. Thus, it is intended that the embodiments cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. An implantable medical device and a system for steering the implantable medical device, the system comprising: an actuation line; a pivot forming a loop coupled to an exterior surface of the implantable medical device; and a tether attached at one end to the actuation line and arranged through the loop of the pivot and configured to orient the implantable medical device in response to tension applied to the actuation line and release from the pivot after the implantable medical device is oriented.
2. The system of claim 1, wherein the loop comprises a layer of loop material that forms a lumen between the exterior surface of the implantable medical device defined by a graft material.
3. The system of claim 2, wherein the actuation line includes an eyelet and the tether is attached to the eyelet at the one end and arranged through the pivot and the eyelet.
4. The system of claim 3, wherein the tether is arranged through the eyelet, arranged through the loop, and subsequently attached to the eyelet.
5. The system of claim 1, wherein the pivot and the actuation line are configured to form a pulley to orient the implantable medical device in response to tension applied to the actuation line.
6. The system of claim 1, further comprising an actuation line sleeve and the tether is pulled into the actuation line sleeve in response to tension applied to the actuation line.
7. The system of claim 6, wherein the actuation line sleeve is attached to the exterior surface of the implantable medical device proximal to the pivot.
8. The system of claim 6, wherein the pivot and the actuation line sleeve are configured to form a pulley to orient the implantable medical device in response to tension applied to the actuation line.
9. The system of claim 1, further comprising a removable lock wire configured to maintain coupling of the tether to the implantable medical device.
10. The system of claim 9, wherein the tether includes an eyelet and the removable lock wire is arranged through the eyelet to couple the removable lock wire to the tether.
11. The system of claim 10, wherein the eyelet of the tether is at or near a proximal end of the tether and a distal end of the tether is coupled to the actuation line.
12. The system of claim 11, wherein the removable lock wire is arranged through a flow lumen of the implantable medical device and the tether is arranged from the flow lumen of the implantable medical device at the proximal end of the tether to an exterior surface of the implantable medical device at the distal end of the tether.
13. The system of claim 12, wherein the tether is configured to release from the actuation line in response to withdraw of the removable lock wire from the eyelet of the tether.
14. The system of claim 9, further comprising a catheter arranged through a lumen of the implantable medical device, wherein a proximal end of the tether is releasably coupled to the catheter, and a distal end of the tether is coupled to the actuation line.
15. A system comprising: a catheter; an implantable medical device arranged near a leading end of the catheter and including a proximal end, a distal end, and a flow lumen extending therebetween; a pivot forming a loop coupled to an exterior surface of the implantable medical device; a sleeve coupled to the exterior surface of the implantable medical device proximal to the loop; an actuation line arranged through a lumen defined by the sleeve; and a tether coupled to the actuation line and arranged through the loop and configured to steer the implantable medical device in response to tension applied to the actuation line; wherein the tether is releasable from the pivot after the implantable medical device is oriented as desired.

16. The system of claim 15, wherein the loop and the sleeve are pivot points and are configured to form a pulley between the tether and the actuation line to steer the implantable medical device in response to tension applied to the actuation line.

17. The system of claim 15, further comprising a removable lock wire configured to maintain coupling of the tether to the implantable medical device, and wherein the tether is configured to release from the actuation line in response to withdraw of the removable lock wire.

18. A method of steering an implantable medical device, the method comprising: delivering the implantable medical device to a target location within a patient's vasculature; and manipulating an actuation line coupled to the implantable medical device by a tether arranged through a pivot forming a loop and coupled to an exterior surface of the implantable medical device to steer the implantable medical device; further including uncoupling the tether from the pivot after the implantable medical device is oriented.

19. The method of claim 18, further including maintaining coupling of the tether to the implantable medical device by a removable lock wire during the steering of the implantable medical device, and releasing the tether from the loop by withdrawing the removable lock wire.
